GNN vs DimeNet for HOMO-LUMO prediction

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Abstract

This project seeks to compare the performance of a baseline Graph Neural Network Model (GNN) with a variant model, DimeNet, which accounts for 3D geometric information and is among the best performing models in the literature for molecule property prediction for e.g. the HOMO-LUMO gap. Model comparisons are evaluated on the QM9 dataset (a dataset containing small organic molecules) and on the HOMO-LUMP gap property prediction task. A study exploring the latent space of the learned models is also attempted using t-SNE. Overall, DimeNet has indeed shown better results than a baseline GNN, highlighting the importance of incorporating 3D geometric information in molecular modeling.

Code Repository

https://github.com/nevinngyt/DeepLearning-CourseProject

1 Introduction

Computational chemistry relies on quantum-mechanical simulations to predict molecular properties such as HOMO/LUMO energies, which are important in evaluating the chemical reactivity, molecular stability, and electronic properties of a molecule. These simulations, typically based on density functional theory (DFT), are accurate but computationally expensive, often requiring hours per molecule. Data-driven surrogate models that predict molecular properties directly from molecular structure can accelerate materials screening, drug discovery, and chemical design by orders of magnitude, allowing researchers to discover molecules that meet their needs faster. The QM9 dataset is one of the most widely used benchmarks for evaluating such models, where it contains approximately 130,000 small organic molecules, each with up to 9 heavy atoms (C, N, O, F) and associated hydrogen atoms. Each molecule is represented as a graph where atoms are nodes and bonds are edges, with full 3D coordinate information for the equilibrium geometry.

2 Methods

Baseline model

The prediction task is on HOMO/LUMO prediction using QM9. A Graph Neural Network (GNN) is chosen as the baseline, given the molecular representation as a graph in QM9 that seems to be more likely for a GNN to be used. GNN is a good choice because it regards nodes as atoms and bonds as edges, which already constitutes the data structure in QM9. The GNN's message passing framework provides good potential for modeling the interactions between the atoms and types of bonds present in the molecule.

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Model variant

A model architecture that uses directional message passing (DimeNet) is explored as a variant. Baseline GNNs represent a molecule as a graph using only the distance between atoms (nodes). However, they do not consider the spatial direction from one atom to another, despite directional information playing a central role in empirical potentials for molecules, e.g. in angular potentials. Hence, directional message passing, in which embedding the messages passed between atoms is done instead of the atoms themselves. Each message is associated with a direction in the coordinate space. These directional message embeddings are rotationally equivariant since the associated directions rotate with the molecule, where these message passing schemes are analogous to belief propagation, which uses the directional information by transforming messages based on the angle between them. Spherical Bessel functions and spherical harmonics were used to construct theoretically well-founded, orthogonal representations that achieve better performance than the currently prevalent Gaussian radial basis representations. This enables graph neural networks to leverage directional information in addition to the interatomic distances that are used by normal GNNs.

3 Results and Discussion

Model evaluation (MAE)

For both the baseline GNN and DimeNet, different model and training configurations were explored. The mean absolute error (MAE) was used as the main evaluation metric (L1Loss) as it is considered the standard evaluation metric. After trial and error, the best performing baseline GNNs have an MAE value of ~0.249 eV, while the best performing DimeNet has an MAE value of ~0.134 eV. From the training loss curves below, DimeNet models reach below 0.1 eV in MAE, unlike baseline GNNs, where their MAE reach much more than 0.1 eV. This demonstrates that the incorporation of 3D geometric information as proposed by DimeNet indeed leads to better performance in the prediction of HOMO/LUMO. Note that training losses for different experiment configurations are logged in Weights and Biases.

The best experiment configuration found so far for baseline GNN is as follows:

- gnn channels: [128, 128]
- pool: "global max pool"
- fc channels: [128, 64, 32]
- activation: "SiLU"

The best experiment configurations found so far for DimeNet are:

- emb size: 128,
- num blocks: 2,
- num bilinear: 8,
- num spherical: 7,
- num radial: 6,
- cutoff: 5.0,
- envelope exponent: 5

Additional analysis: t-SNE

For the baseline GNN, it can be seen that the embeddings clusters are jumbled together, and the color transitions are blocky and abrupt. For DimeNet, the embeddings are smoother and have a more continuous gradient. This indicates that DimeNet has learned a more continuous mathematical manifold that accurately maps to the underlying physics of the molecules. It can also be observed that the latent spaces for baseline GNN often collapse into single blobs, whereas for DimeNet, the latent space can be observed to break apart into distinct "islands". This may be due to DimeNet being more sensitive to exact geometries and thus spontaneously clustering molecules by chemical family, even though the model is not explicitly trained to classify functional groups.

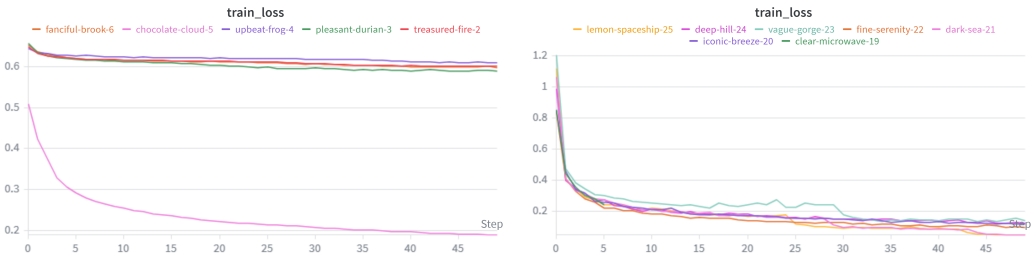

Figure 1: *Left*: Training loss for baseline GNN. *Left*: Training loss for DimeNet

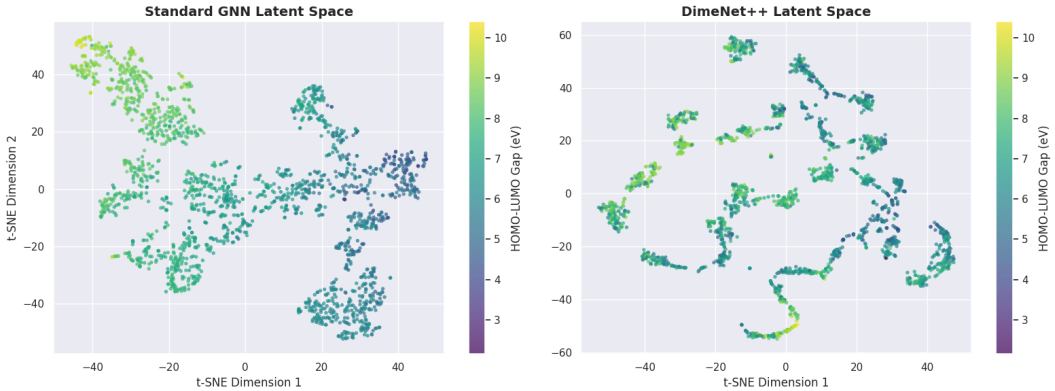

Figure 2: t-SNE visualisation for baseline GNN vs DimeNet

4 Conclusion

Overall, this project demonstrates that Directional message passing enables graph neural networks to leverage directional information in addition to the interatomic distances that are used by normal GNNs, which results in better performances in molecular property prediction, in this case for HOMO/LUMO energies. These can be observed by the lower MAE values and smoother/more well-separated latent space representations in DimeNet. Given more time/compute, other datasets like MD17, other molecular properties (18 other properties like dipole moment, free energy, etc.), other variant architectures like SchNet and more experiment/training configurations could be explored to see how the performances vary as a result.

Collaboration Statement

AI was used for code debugging, understanding code from literature, and understanding documentation libraries for e.g. torch-geometric.

References

Gasteiger, J., Groß, J. Günnemann, S. (2020). Directional message passing for molecular graphs. International Conference on Learning Representations (ICLR).

PyG item (PyTorch Geometric) https://PyTorch Geometric.readthedocs.io/en/2.7.0/index.html